

Clusterless Decoding of Monkey Decisions from Neural Spikes data in iterative Battle of Sexes

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1. Introduction

The decoding of neural activity to predict decision-making processes in non-human primates has been a subject of great interest in neuroscience. In this report, we present a study focused on training a decoder to predict the decision outcomes of monkeys and their opponents during an interactive behavioral task called iterative battle of sexes (iBOS). We investigate the decoder’s performance in predicting the monkey’s decision, the opponent’s decision, coordination or non-coordination between the monkeys, and the final outcome.

2. Related Work

A related work to this study is Neuronal prediction of opponent’s behavior during cooperative social interchange in primates [1]. This study examines the neuronal basis of social prediction and its influence on cooperative interactions. Specifically, specific neurons in the primate dorsal anterior cingulate are identified as being involved in predicting an opponent’s unknown decision to cooperate or defect in a mutually beneficial situation.

these other-predictive neurons in facilitating cooperative behavior and propose potential implications for the targeted treatment of social behavioral disorders. In our study, we will be using a similar experiment to the one used in [1] (iterative Prisoner’s Dilemma) which is iBOS: iterative Battle of sexes. Different rewards will be given, but a similar experiment flow will be used (Figure 1).

Another related work is Decoding and perturbing decision states in real time [2]. This study utilizes real-time neural feedback control of stimulus duration and reveals a strong correlation between motor cortex-decoded Decision Variable (DV) fluctuations and decision states in macaques. These fluctuations outperform average DV and visual stimulus alone in predicting behavioral choices. Additionally, observed robust changes in DV sign align with patterns seen in studies on changes of mind.

3. Method

3.1. Data Preprocessing

The data utilized in this study consists of a sequence of 96x1 vectors representing preprocessed spike data. To preprocess the data, we apply a half-Gaussian filter, which helps to enhance the relevant features of the neural activity. (Fig. 1)

Diverse data preprocessing methods were considered such as:

- preprocessed using the half of a Gaussian with standard deviation ν seconds for grouping the spikes.
- preprocessed by counting the number of spikes in a window of Δ milliseconds.
- preprocessed by averaging the number of spikes in a window of Δ milliseconds.

We tested these different methods and chose the one that leads to the best decoder experience.

Moreover, the decisions are either selfish (0) or altruistic (1).

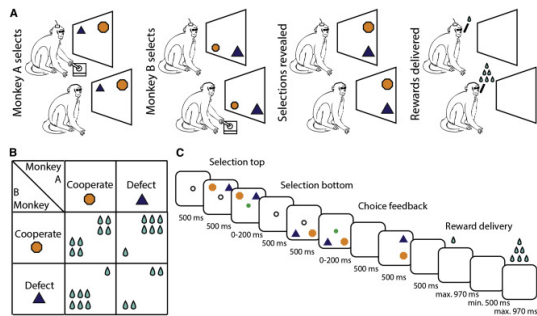


Figure 1. (A) Experimental set-up. (B) Payoff matrix. (C) Trial timeline.

Additionally, the researchers uncover distinct neurons that encode the monkey’s own decision based on previous outcomes. These findings underscore the significance of

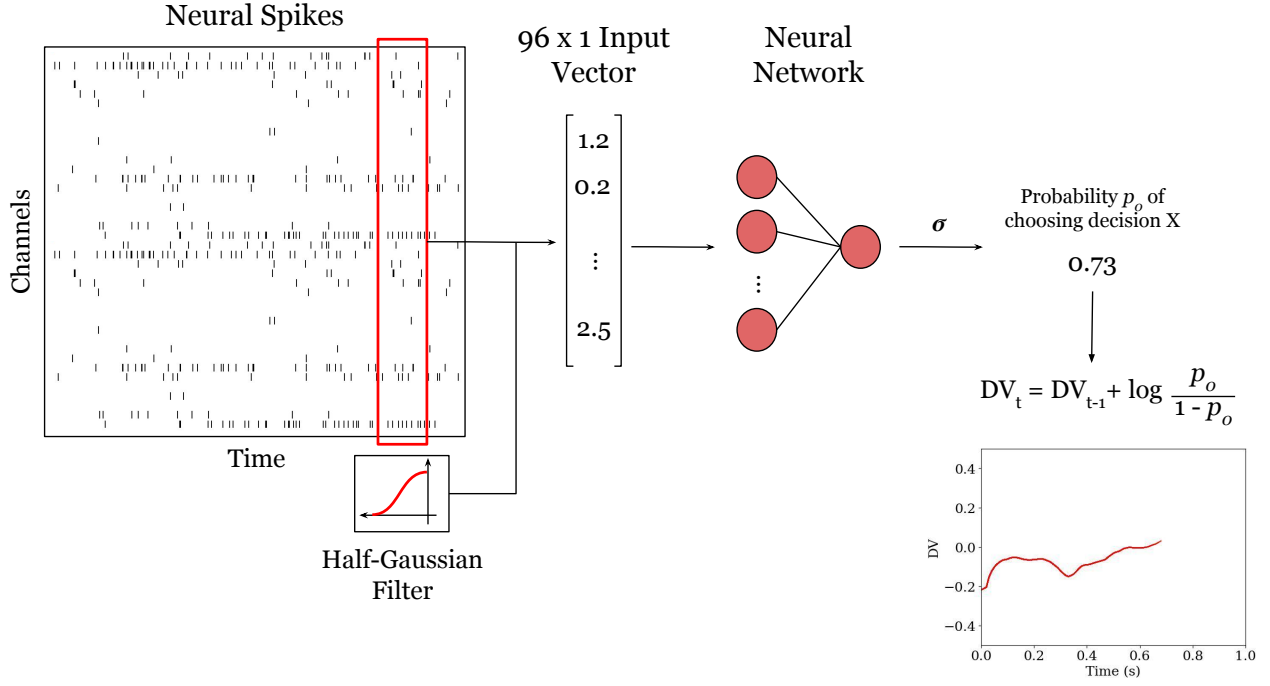


Figure 2. Data preprocessing, clusterless decoding and decision variable update. [1]

3.2. Decoder Architecture

The decoder architecture employed in this study consists of 96 parameters, corresponding to the dimensionality of the input data (Figure 2). The model is regularized with an L1 regularization. The primary objective of the decoder is to accurately predict the decision-related variables of interest. We utilize a learning rate of 0.05 during the training process, which spans 100 epochs. We normalize the training data by computing the mean μ and standard deviation σ of the training set. We use and update μ and σ to normalize the test set at test time in the following way:

$$\mu_{n+1}^{(test)} = \frac{n\mu_n^{(test)} + x_{n+1}^{(test)}}{n+1}$$

$$\sigma_{n+1}^{(test)} \approx \sqrt{\frac{(n-1)\sigma_n^{(test)2} + (x_{n+1}^{(test)} - \mu_{n+1}^{(test)})^2}{n}}$$

Similarly to the data preprocessing we explored diverse decoder architectures. We tried kNN (k Nearest Neighbors) Decision trees, random forest, ensemble model of linear regression models and a 2 layer neural network. After testing these different methods, we found out that using a simple linear regression was the model leading to best performance. In fact, it has a good enough complexity to capture the patterns in the data and yet not overfit the data.

3.3. Training the model

We split the data into a training, validation and testing set. We fine-tune the hyperparameters using Optuna [3] on the validation set and test our model using the test set. We optimize the parameters of the model using Adam with learning and number of epochs being hyperparameters. The loss function is binary cross-entropy:

$$\mathcal{L} = \frac{1}{n} \sum_{i=1}^n y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)$$

3.4. Hyperparameters Selection

Hyperparameter	Value
α	0.05
n_{epochs}	100
λ	0.01
δ	0.05
Δ	0.3

Table 1. Hyperparameters found using Optuna.

In this model, we have multiple hyperparameters like the learning rate α , the number of epochs n_{epochs} for training and the regularization term λ , the duration on which the Gaussian filter is applied Δ , the time difference between

different data vectors δ . These parameters are optimized using Optuna [3] which is a hyperparameter optimization framework to automate hyperparameter search. This is the hyperparameters table that we found (Table 1).

3.5. Evaluation

To evaluate the model, we look at the balanced accuracy of the predicted values compared to the ground truth. We also test the model in a context of real time decoding of the decision. In this case, we define a decision variable in the following way:

$$DV_t = 2\sigma\left(\sigma^{-1}\left(\frac{DV_{t-1} + 1}{2}\right) + \log \frac{p_0}{1 - p_0}\right) - 1,$$

where σ is the sigmoid function and p_0 is the probability of the decision being 1 or altruistic decision.

This decision variable will take a value between -1 and 1. Thus, to evaluate the model, we look at the sign of the decision variable after 3 seconds or when the decision variable crosses a certain threshold and compare the outcome to the ground truth. Note that in this case, time 0 corresponds to fixation cue on event.

4. Results

4.1. Decoder performance on one session

	First	Second	Total	Random
Self Decision	77%	72%	75%	50 %
Other Decision	66%	71%	69%	50 %
Coordination	62%	66%	64%	50 %
Outcome	52%	56%	54%	25 %

Table 2. Decoding accuracy of various decision-related variables by training a decoder on 10 trials and testing on 378 trials.

Using the specified decoder architecture and data preprocessing techniques, we employ the decoder to predict the following variables: the monkey’s decision, the opponent’s decision, the final outcome, and the coordination or non-coordination between the monkeys. For training, we focus on the last 1.5 seconds of each trial, as this period exhibits the highest confidence in terms of its association with the decision-making process. Our evaluation is based on training the decoder on 10 trials and subsequently testing it on 378 trials. The decoding accuracy results are summarized in Table 2.

From the results, it is evident that the decoder exhibits the highest accuracy in predicting the monkey’s own decision. Notably, when the monkey plays as the second participant, the decoder performs better in predicting the decision of the other monkey. However, the decoder struggles

to accurately predict the coordination or non-coordination outcome. In the following figure, we understand why this happens:

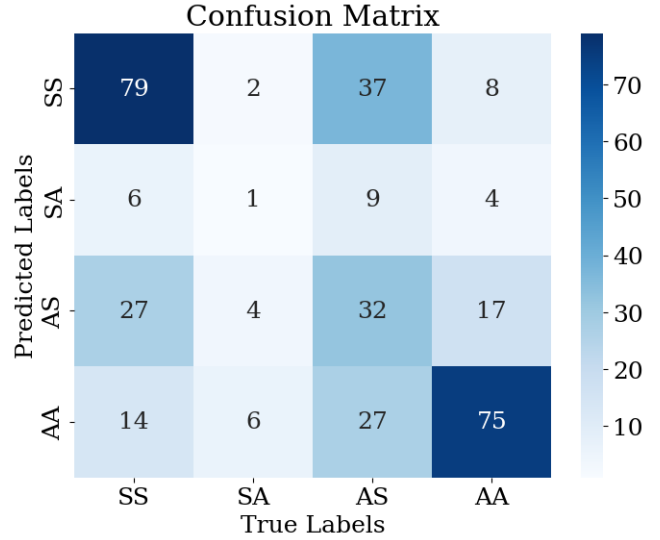


Figure 3. Confusion matrix of the decoder predicting the outcome. SS, AS, SA, and AA correspond to the decision of the self and other monkeys, with S for selfish and A for altruistic.

We observe that when computing the accuracy of coordination and non-coordination, we have to sum up the two diagonals of this confusion matrix. Since the sum of the second diagonal isn’t large, this explains why prediction coordination and non-coordination leads to worse results than predicting the monkey’s decision or the other monkey’s decision.

To further analyze the performance, we compare the accuracy of the decoder when trained on data from trials where the monkey played as the first participant versus the second participant. Surprisingly, both cases yield similar accuracy rates, as shown in Table 3.

Trained on First	72%
Trained on Second	70%

Table 3. Decoding accuracy of the monkey’s (self) decision by training a decoder on 10 trials and testing on 378 trials.

To assess the impact of the number of trials and the number of samples per trial on the decoder’s performance, we conduct additional experiments. Figure 4 depicts the decoding accuracy of the monkey’s decision and the other monkey’s decision, respectively, as a function of the number of samples used per trial and the number of trials employed for training. Due to the use of regularization, we observe that the accuracy increases but can’t reach higher values due to

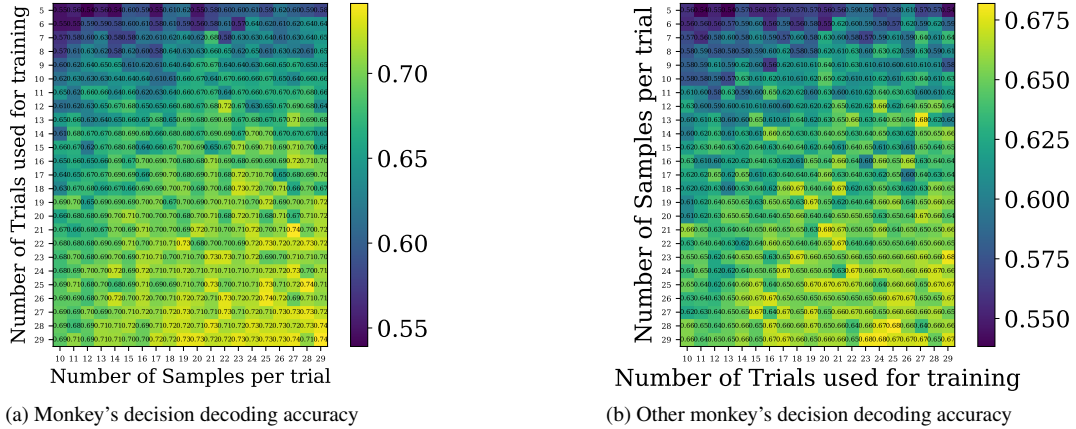


Figure 4. Decoding accuracy as a function of samples per trial and training trials for a single session.

the small number of data points in the session. We observe that using 20 trials and 14 samples per trial, we can have an accuracy above 70%.

4.1.1 Decoding monkey's self-decision

As we saw above, the decoder is able to best decode the monkey's own decision. Let's evaluate how good is this decoder to do closed loop experiments. For the closed loop experiment, we decide to either stop the trial when the decision variable exceeds a certain threshold or when the fixation cue has been on 3 seconds ago.

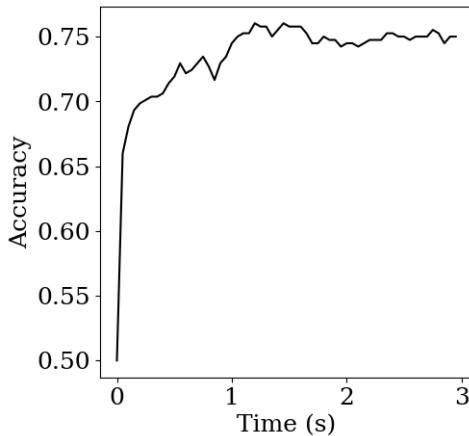


Figure 5. Accuracy of decoding the monkey's decision from one session as a function of time.

We plot the accuracy of this decoder as a function of time and observe the results in Figure 5. We see that as time increases the performance of the decoder increases which means that the decision variable we are using is a reliable

metric to decode the monkey's decision.

On another hand, we take a look at the relation between the accuracy and the threshold chosen for the decision variable. We expect that the higher the threshold chosen, the better the accuracy. We plot the accuracy as a function of the threshold and observe the graph in figure 6. As expected, we see that the accuracy increases with the threshold value.

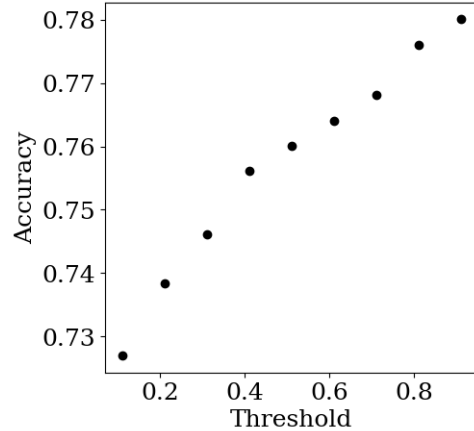


Figure 6. Accuracy of decoding the monkey's decision from one session as a function of threshold.

Now, we take a look at the decision variable as a function of time (Figure 7). We observe that the decision variable reaches high values faster when the monkey's decision is selfish (0) than when it is altruistic (1). This might mean that selfish decisions are easier to take for the monkey while altruistic decisions are harder. Moreover, we observe a clear distinction between the red and blue curves in the Figure 7 after a few milliseconds seconds after Fixation Cue on.

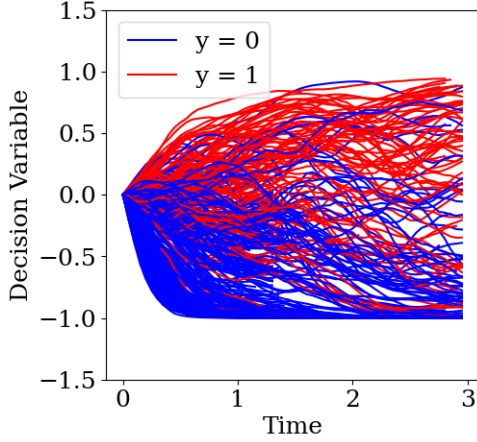


Figure 7. Decision Variable as a function of time. Red and blue curves correspond to altruistic and selfish decisions respectively.

4.1.2 Change of Minds (CoM)

In the following section, we take a look at the changes of minds observed in the decision variable plot. We assume that our decoder has a good enough performance to decode the monkey’s decision in real time based on our previous discussions. We plot the decision variable curves that present a change of mind behavior, where the decision was either altruistic or selfish and then changed at the end of the decision time. We observe 76 CoM in this session of 388 trials. The proportion of selfish and altruistic decisions that encompasses CoM is the same as in the session about 33% altruistic and 67% selfish. (Figure 8)

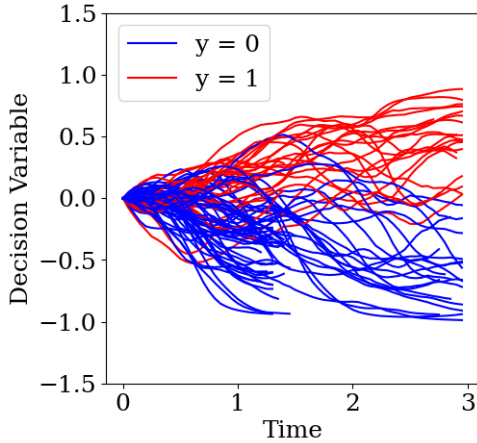


Figure 8. Decision Variable showing CoM as a function of time. Red and blue curves correspond to altruistic and selfish decisions respectively.

Now, let’s take a look where did these CoM happen in a session (Figure 9). We observe that these CoM would

happen in waves where we have periods of time where the monkey has these changes of mind happening and then periods of time where it is sure what it wants. This might be moments of doubts where the monkey considers changing its strategy.

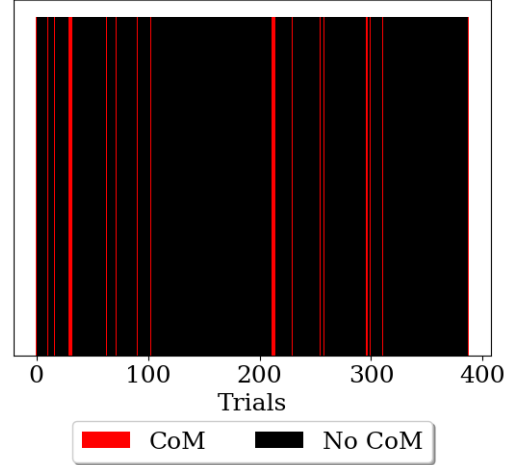


Figure 9. Trials showing changes of mind during a session. Red and black strips correspond to trials with and without changes of mind respectively.

4.2. Decoder performance on other sessions

We aim to predict the monkey’s decision by training our model on one session and testing it on another. However, we have observed that the accuracy remains around 50%. As a result, the model fails to generalize well to different sessions. To address this issue, a potential solution is to construct a comprehensive dataset comprising numerous sessions and train a model using this data. By doing so, we anticipate that the resulting model will exhibit better transferability across a wider range of sessions.

5. Conclusion

In conclusion, we investigated the decoding of neural activity to predict decision-making processes in primates during an interactive behavioral task called iterative battle of sexes (iBOS). We trained a decoder to predict various decision-related variables, including the monkey’s own decision, the opponent’s decision, coordination or non-coordination between monkeys, and the final outcome.

Our results indicate that the decoder performs best in predicting the monkey’s own decision, achieving an accuracy of 75%. However, it struggles to accurately predict coordination outcomes and exhibits lower accuracy in predicting the opponent’s decision and the final outcome.

We explored different data preprocessing techniques and decoder architectures, finding that a linear regression model

yielded the best performance. We also conducted experiments to evaluate the impact of the number of trials and the number of samples per trial on the decoder’s performance. We observed that using 20 trials and 14 samples per trial resulted in an accuracy above 70

Additionally, we conducted closed-loop experiments and found that the decoder’s accuracy increased over time, indicating the reliability of the decision variable in decoding the monkey’s decision. We also observed changes of mind (CoM) in the decision variable, where the decision switched from altruistic to selfish or vice versa during the decision time.

However, when testing the decoder on different sessions, the accuracy dropped to around 50%, indicating a lack of generalization. To improve generalization, it would be beneficial to construct a comprehensive dataset comprising multiple sessions and train a model using this data.

References

- [1] Keren Haroush and Ziv M Williams. Neuronal prediction of opponent’s behavior during cooperative social interchange in primates. *Cell*, 160(6):1233–1245, 2015. [1](#), [2](#)
- [2] Diogo Peixoto, Jessica R Verhein, Roozbeh Kiani, Jonathan C Kao, Paul Nuyujukian, Chandramouli Chandrasekaran, Julian Brown, Sania Fong, Stephen I Ryu, Krishna V Shenoy, et al. Decoding and perturbing decision states in real time. *Nature*, 591(7851):604–609, 2021. [1](#)
- [3] Optuna. Optuna, 2017. <https://optuna.org/>. [2](#), [3](#)